

Artificial Intelligence

Homework 9

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November 19, 2025

Deadline

December 3, 2025

Because we will have the Thanksgiving break, I extend the time for this homework to 3 weeks (instead of 2 weeks as usual). The deadline is December 3, 2025.

1 AI for Software Development via Prompt Engineering - 30 Points

In this assignment, you will use Large Language Models (LLMs) like ChatGPT or Gemini to construct a working implementation of an Evolutionary Algorithm (EA) for solving the MAX-ONE problem. The goal of the MAX-ONE problem is to maximize the number of '1's in a binary string. You will use AI-generated code to implement core components of the EA, including functions for fitness computation, crossover, mutation, and selection.

During one of our lectures, I have implemented an EA implementation for the MAX-ONE problem. The Python code is available in files `MAXONE_EA_2024.py` or `MAXONE_EA_2025.py` in folder `Code` on Canvas. However, the objective of this assignment is to use AI to generate the code and you, as a software engineer, have to test the correctness of the AI-generated code. You should make sure that the AI-generated code returns reasonably good result as the human's code. You can use either ChatGPT or Gemini. You have to report your prompts and your AI-generated code.

Your tasks are as follows.

1. Make sure that the LLM understands the MAX-ONE problem by giving it the context.
2. Make sure that the LLM understands the overall, high-level description of Evolutionary Algorithm.
3. Utilize the LLM to generate code snippets for the components of an Evolutionary Algorithm. This will include functions for:
 - Fitness computation
 - Crossover
 - Mutation
 - Selection

You must write clear and precise prompts for the LLM, specifying the expected functionality and API of each function.

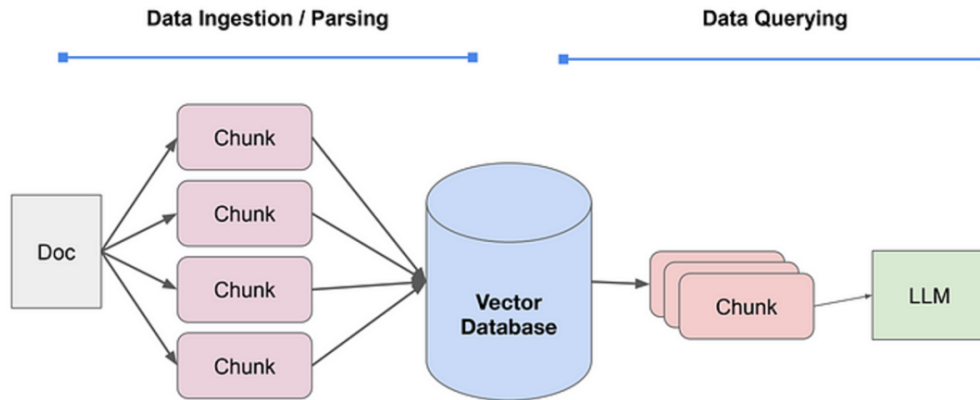


Figure 1: A Retrieval-Augmented Generation (RAG) System.

4. Once you have generated code for each component, you can use the LLM again to help you combine the individual functions into a fully working implementation of the Evolutionary Algorithm.
5. You must test your implementation on the MAX-ONE problem. Make sure the algorithm works correctly and efficiently converges towards the optimal solution.

2 Building a simple RAG system - 70 Points

Objective

In this assignment, you will build a **Retrieval-Augmented Generation (RAG)** system (see Figure 1) using the dataset provided, which contains information on data scientists (see file `data_scientist_salaries.csv` in the folder **Homework** on Canvas). The goal is to leverage this dataset to answer questions and provide relevant insights using a combination of a retriever and a generative AI model. You should use the free Gemini API.

Dataset Overview

The CSV file, `data_scientist_salaries.csv` in the folder **Homework** on Canvas, you have been provided contains information related to data scientists, such as their job titles, salaries, experience levels, locations, and other relevant details. Your task is to utilize this data to build a RAG system that can intelligently retrieve and generate answers based on the information in the dataset.

Task Requirements

1. Understanding the RAG Architecture:

- A RAG system (see Figure 1) combines a retrieval mechanism with a generative model. The retriever is used to fetch relevant pieces of information from a dataset or knowledge base, which the generative model then uses to produce a coherent response.
- Your RAG system will consist of two main components:
 - **Retriever:** This component retrieves the most relevant entries from the dataset based on a query.
 - **Generator:** This component generates a detailed and contextual response based on the retrieved data.

2. Using LLM for the Generative Component:

- You will use an LLM (i.e., Gemini) to generate responses. The retriever should first narrow down the relevant information from the dataset, which will then be passed to the LLM to generate a detailed answer.
- The system should be able to answer questions such as:
 - “What is the average salary of data scientists with over 5 years of experience?”
 - “What are the top locations for data scientist jobs?”
 - “How does experience level impact salaries for data scientists?”

3. Implementing the Retriever:

- You need to create a function that efficiently retrieves relevant rows from the CSV dataset based on the user’s query.
- Consider using techniques such as keyword matching, semantic similarity, or embeddings to find the most relevant entries in the dataset.
- You can use libraries like `pandas`, or `scikit-learn` to assist with data retrieval.

4. Integrating the Generator:

- Use the retrieved data to provide context for your LLM prompt, allowing it to generate an accurate and insightful response.
- Write clear and structured prompts to ensure the LLM utilizes the retrieved information effectively. For example:
 - **Prompt:** “Using the data provided, summarize the trends in data scientist salaries based on experience level.”
 - **Context (retrieved data):** A subset of the dataset containing information on salaries segmented by experience level.

5. Testing and Evaluation:

- Test your RAG system on various types of queries to ensure it retrieves and generates responses accurately.
- Analyze how changes to the retrieval method impact the quality of the generated answers.
- Ensure that your system is efficient in both retrieving relevant data and generating coherent responses.

Deliverables

1. **Code Implementation:** Submit a Python script or Jupyter notebook containing your RAG system implementation. Your code should be well-structured, modular, and thoroughly commented.
2. **Report:** Write a brief report (1-2 pages) explaining the following.
 - How you implemented the retrieval mechanism.
 - The prompts used for the LLM and how you integrated the retrieved data into the prompts.
 - Examples of queries and the system’s responses.
 - Any challenges you faced and how you overcame them.

Grading Criteria

- Correctness of the RAG system implementation (40%)
- Effectiveness of the retrieval and generation components (30%)
- Code quality and readability (20%)
- Report clarity and completeness (10%)

Tips for Success

- Be clear and specific when writing prompts for the LLM to ensure accurate responses.
- Optimize your retrieval process to minimize latency and maximize the relevance of the retrieved data.
- Experiment with different methods for query processing and retrieval to improve the quality of your system.
- There are many columns / features in the dataset. For the data preprocessing, you can keep the most relevant and important columns such as Hobby, OpenSource, Country, Student, Employment, FormalEducation, UndergradMajor, CompanySize, DevType, YearsCoding, Salary, SalaryType, ConvertedSalary.